

Loan Default Risk Analysis

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Task 7 – Critical Evaluation of Principal Component Analysis (PCA)

PCA Advantages and Limitations

Advantages	Limitations
Reduces the number of variables.	Original variables become harder to understand.
Reduces repeated information from correlated variables.	Mainly suitable for numerical variables.
Can reduce model complexity.	PCA is not always the best method.
Can keep most important information using fewer components.	The result depends on the dataset.
Our PCA: first 5 components explain 73.88% of the variance.	Our dataset has only 10 numerical variables, so the need for PCA is limited.
One reviewed paper reduced 159 variables to 9 components while keeping 84.3% of the variance.	Results from other datasets cannot be directly applied to our dataset.

Final Critical Evaluation

The reviewed papers show that PCA can reduce the number of variables and repeated information. However, PCA should not be used automatically because its usefulness depends on the dataset. In our Loan Default dataset, PCA was applied to 10 numerical variables. The first 5 principal components explain 73.88% of the total variance, while the first 7 components explain 92.17%. This shows that PCA can reduce the number of variables while keeping most of the information. However, our dataset has only 10 numerical variables, so PCA is not necessarily required. Therefore, PCA can be considered as an optional technique for our project.

PCA Results

Selected numerical variables: 10

First 5 PCs – cumulative variance: 73.88%

First 7 PCs – cumulative variance: 92.17%